

IoT Based Smart Greenhouse Design with an Intelligent Supervisory Fuzzy Optimized Controller

Mh. Aghaseyedabdollah*, Y. Alaviyan, A. Yazdizadeh

Department of Control Engineering, Shahid Beheshti University, Tehran, Iran
 m.aghaseyedabdollah@mail.sbu.ac.ir, y.alaviyan@mail.sbu.ac.ir, a_yazdizadeh@sbu.ac.ir

Abstract— Internet of things (IoT) alludes to the network of physical objects that are connected to the internet, to share data and send control commands. It has been used in many plants for different purposes. This paper presents an intelligent supervisory fuzzy controller (ISFC) to control the temperature, soil moisture, and humidity in the smart greenhouse by using IoT. The introduced ISFC checks data and prevents plant damage and also has an alarming system to notify users. Membership functions of the proposed controller are optimized by using the Jaya algorithm. It has been designed to be users friendly so that the user can monitor and change the desired value of the parameters of the greenhouse remotely and also be notified of the occurrence of any event like a fire. Practical results show that greenhouse parameters are successfully controlled by the suggested controller.

Keywords—Smart Greenhouse, Internet Of Things, Jaya Algorithm, Supervisory Control, Intelligent Control.

I. INTRODUCTION

The network of physical objects that are connected to the internet which collecting and sharing data, refers to the internet of things (IoT). The IoT make the environment, homes, offices, and vehicles smarter and efficient [1]. In traditional greenhouses, manual control makes reduces production, energy loss, and less effective. In a smart greenhouse, sensors measure the value of the parameter then controller sends the appropriate command to actuators according to desired and measured values. This structure leads greenhouse to optimize production and enhancing product quality [2].

In [5] PID controllers are suggested on the basis of fusion of different parameters to energy consumption and in [6] optimal controllers can be described to minimize energy input that cooling, Heating, and the amount of CO₂ were used as control variables. There are several benefits of using IoT technology in the greenhouse such as reduced wiring, maintenance cost. In order to control and monitor the greenhouse parameters without human intervention, the use of IoT technology and online monitoring in greenhouse structure is suggested [3].

In IoT-based smart greenhouse, data are sent to the cloud for more inspection. To adjust greenhouse parameters several control strategies have been suggested. In [4], the authors have proposed a model predictive controller for accurate irrigation application in the smart greenhouse.

The proposed method can reduce water utilization by about 30% over a period of 3 weeks. Should be noted that nonlinear behaviors of greenhouse and external environmental parameters reduced the control performance of linear control methods therefore intelligent methods are utilized to overcome these issues. In recent years fuzzy logic controller (FLC), adaptive neural fuzzy inference system

(ANFIS) and artificial neural networks (ANN) is suggested to overcome some issues that conventional control techniques. can't have good performance. In [7] a convolutional neural network a technique for monitor the growth of lettuce are designed in agriculture application. In [8] by using the ANFIS model for prediction plant seeds and process the data with high difficulty and purvey good divination results. The fuzzy logic method has a simple structure and fast computation time compared to neural networks that are intricate structures. FLC is a good choice for the control of greenhouses and much research is proposed [9, 10]. In automatic irrigation with measured the soil depth and surface humidity, value of water required tuned by the fuzzy controller [11, 12].

For the purpose of design an FLC, the experience or science of a specialist is needed to generate control rules and express membership functions. For better control performance, optimization methods are utilized to tune the parameters of FLC. There are several methods to tune the parameters of FLC. One of the powerful metaheuristic algorithms is the Jaya algorithm that needs just the general control parameters for its working and does not require any specific parameters. In [13] by utilizing the Jaya algorithm optimized fuzzy controller parameters.

In this paper, an intelligent supervisory fuzzy controller (ISFC) is present to control the temperature, soil moisture, and humidity of an IoT-based laboratory smart greenhouse. ISFC checks data and prevents plant damage and also has a fire alarm system to notify users. Membership functions of the FLC are optimized by using the Jaya algorithm. Also, this possibility is provided for users to monitor and change the desired value of parameters of the greenhouse remotely online.

This paper formed as follows: in section II, the structure of the greenhouse will be presented. The process of our controller will be explained in section III. In section IV we express experimental results. The conclusion will be illustrated in section V.

II. DESIGN AND CONTROL PROCESS OF THE PROPOSED SMART GREENHOUSES

A. Greenhouse physical set up

In this section, the main structure of the greenhouse and intelligent control is explained. Temperature, humidity, and soil moisture are the environmental greenhouse parameters. Mentioned parameters are controlling by an intelligent supervisory optimal Jaya-based fuzzy controller. The manufactured smart greenhouse is shown in Fig. 1. Schematic IoT communication, main sensors and actuators are used in our greenhouse are shown in Fig. 2.

Mega 2560 Arduino board and for Wi-Fi module, the Esp8266 is utilized in the control structure

III. INTELLIGENT CONTROL OF THE GREENHOUSE PARAMETERS

In Fig. 3, greenhouse parameters control process is explained. In order to prevent damage to the plant in different condition, the supervisor controller is used to applying the essential commands to the actuators. The proposed FLC receive data from the sensors and based on generated fuzzy rules, appropriate command sends to actuators for controlling environmental parameters in the greenhouse.

The performance structure of the supervisory controller is shown in Fig. 4. The aim of limitation in this figure is to keep values of greenhouse parameters in a bound limit in order to greenhouse safe operation in environmental conditions.

In this research, the Jaya algorithm is tuning the membership function of the fuzzy controller for improving control performance. Jaya algorithm is a strong optimization algorithm for accomplishing unconstrained and constrained optimization problems. The goal of the proposed method is referred to the idea that the result acquired for a certain issue towards the best solution and far from the worst result. The main objective of this method is fewer iterations and short computational time for converges to the globally optimal values. In Fig. 5 the flowchart of the proposed Jaya algorithm is shown

A. Design and implementation of the temperature controllers

In this part, the environmental temperature of the greenhouse is controlled. The LM35 sensor is measures temperature quantity. Fan and heater are actuators of the temperature control system. The suggested fuzzy controller has a two-input and two-output system. The first input of the

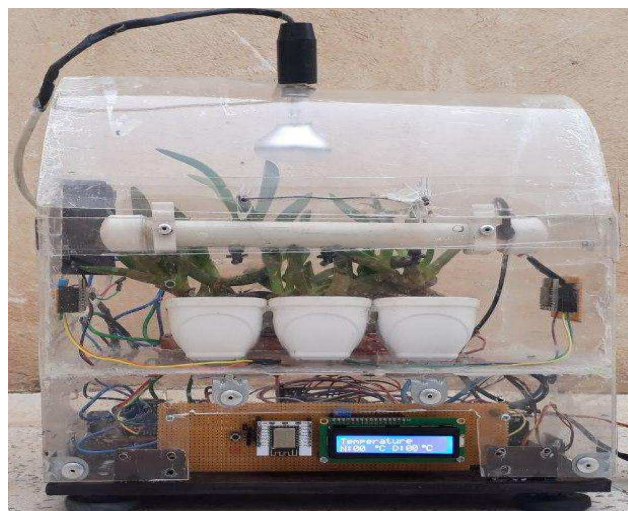


Fig. 1. Smart greenhouse



Fig. 2. Greenhouse structure schematic

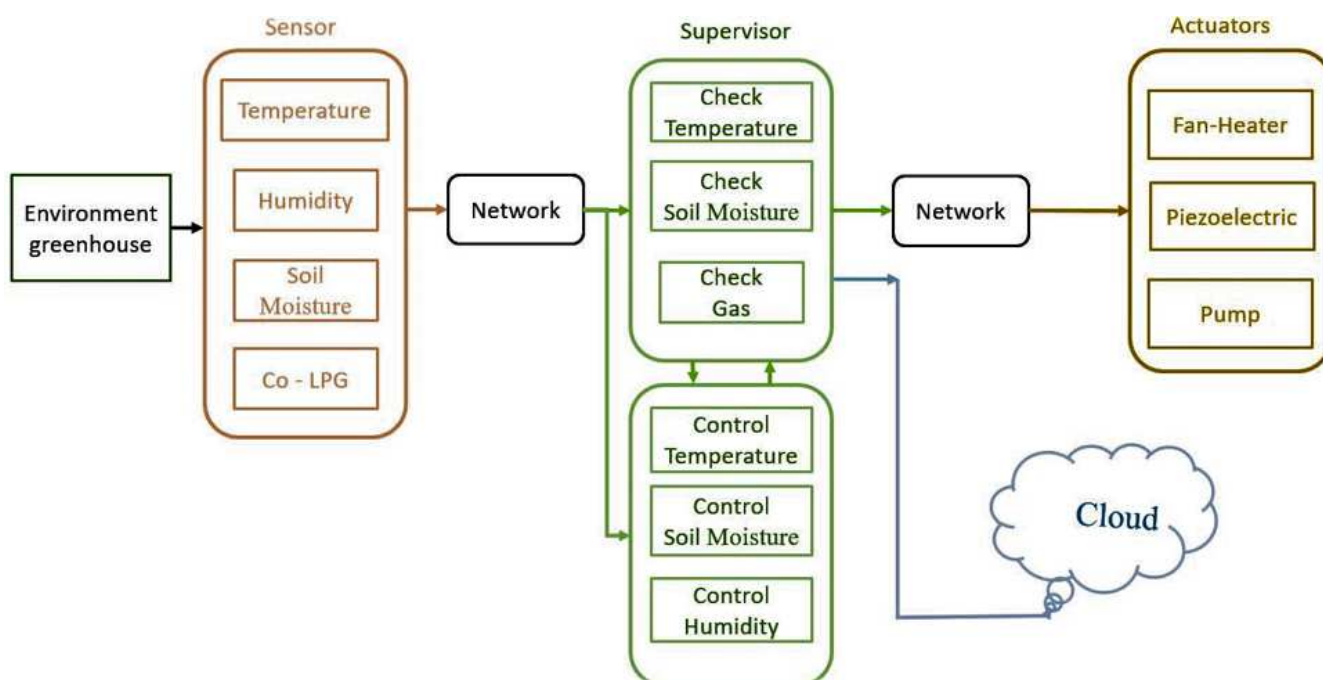


Fig. 3. The control structure of greenhouse parameters

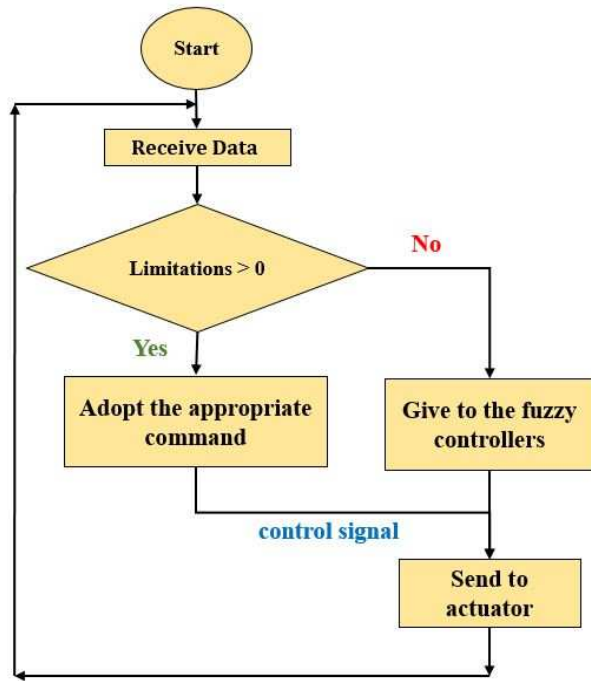


Fig. 4. Performance structure of the supervisory controller

fuzzy controller is the error which difference between the greenhouse temperature and the optimal temperature and the second input is the rate of error. The first output is a fan and the second output is the heater. The supervisor controller chooses which actuator is turn on to prevent to damage the plant. Fig. 5. represents temperature membership functions of the input and output. The fuzzy rules comprehensive in Table I. The defined fuzzy variables for the suggested fuzzy controller are PB (Positive Big), NB (Negative Big), NS (Negative Small), PS (Positive Small), and Z(Zero). The input and output membership functions of temperature control is illustrated in Fig.6.

In Table II change of the value of temperature when the heater is turned on is shown, and also the change of the value of temperature when the fan is turned on are shown in Table III. Also, the proposed tables contain the Nominal power percentage of the heater and fan for tuning the temperature.

B. Design and implementation of the soil moisture controller

In this part, the soil moisture of the greenhouse is controlled. The Y1-69 sensor measures soil moisture value.

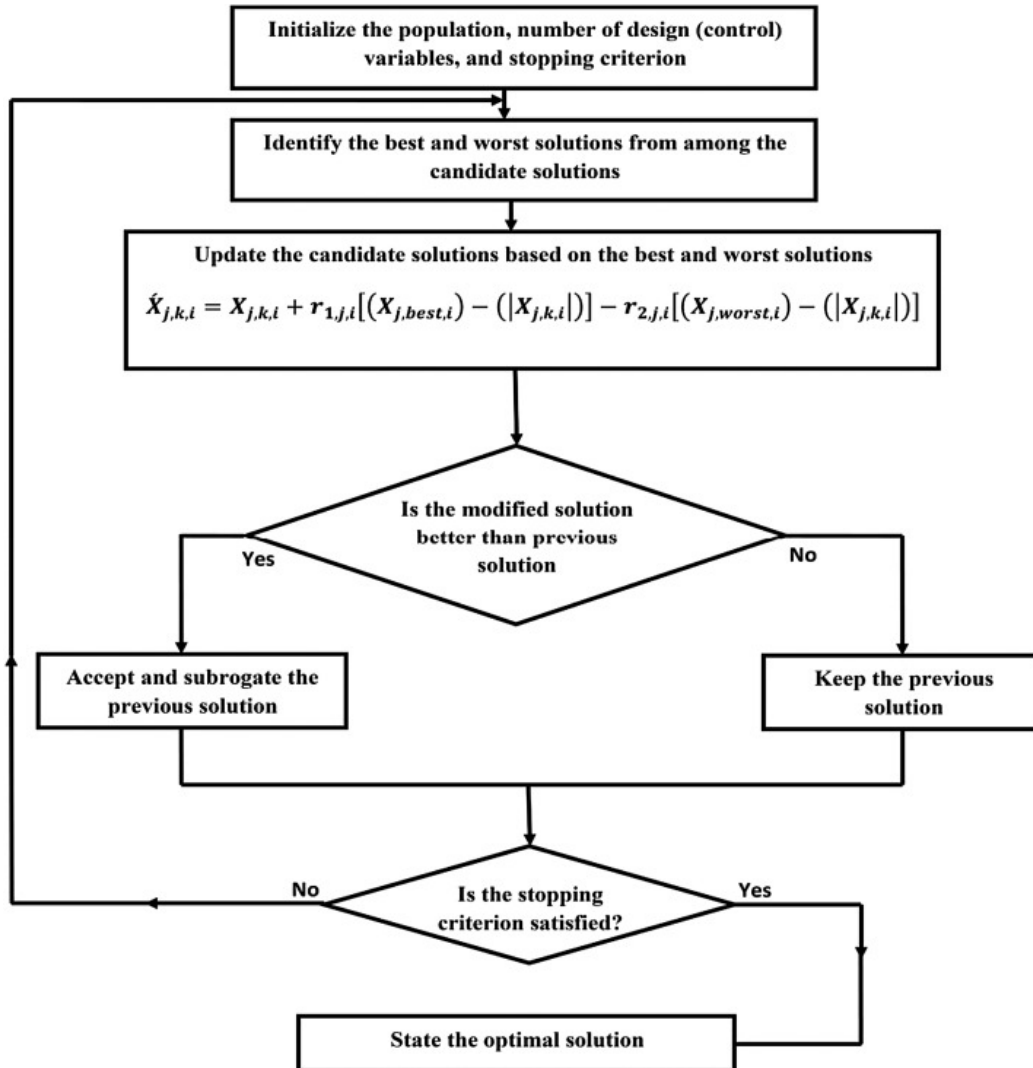


Fig. 5. Flowchart of the Jaya algorithm [14]

The pump is the actuator of the soil moisture control system. The suggested fuzzy controller has the one-input and one-output system. The error is the input of the fuzzy controller which the difference between the greenhouse soil moisture and the optimal soil moisture and on/off is the output. Fig. 7. represents soil moisture membership functions of the input and output. The fuzzy rules comprehensive in Table IV. In Table V change of soil moisture's value when the pump is on and also, the proposed tables contain the time that pump is on for tuning the soil moisture.

C. Design and Implementation of the Environmental Humidity Controller

In this part, the humidity of the greenhouse is controlled. The Dht12 sensor measures humidity value. The manufactured humidity control system is shown in Fig. 8. The piezoelectric is the actuator of the humidity control system. The suggested fuzzy controller has a two-input and one-output system. The error and rate of error are the input

TABLE I. TEMPERATURE CONTROL FUZZY RULES

$\begin{matrix} \dot{e} \\ e \end{matrix}$	NB	NS	Z	PS	PB
NB	M	B	VB	B	M
NS	S	M	B	M	S
Z	VS	S	M	S	VS
PS	S	M	B	M	S
PB	M	B	VB	B	M

TABLE II. VALUE OF TEMPERATURE (HEATER)

$\begin{matrix} \text{Nominal power} \\ \text{Time} \end{matrix}$	%25	%50	%75	%100
0s	25	25.5	25.2	25.1
30s	25.4	26	26.5	26.9
60s	25.8	26.5	27.4	27.8
90s	26.1	27	27.9	28.8
120s	26.2	27.3	28.6	29.7
150s	26.2	27.6	29	30.6
180s	26.3	27.8	30	31.5

TABLE III. VALUE OF TEMPERATURE (FAN)

$\begin{matrix} \text{Nominal power} \\ \text{Time} \end{matrix}$	%25	%50	%75	%100
0s	42.8	42.2	41.9	42
30s	40.5	37.8	38.1	36.4
60s	38.4	35.5	35.3	32.2
90s	36.7	34	33.7	30.9
120s	35.6	33	32.6	30.2
150s	34.8	32.3	31.7	29.7
180s	34.2	32.2	31.1	29.4

TABLE IV. SOIL MOISTURE CONTROL FUZZY RULES

e	PUMP
N	OFF
Z	OFF
P	ON

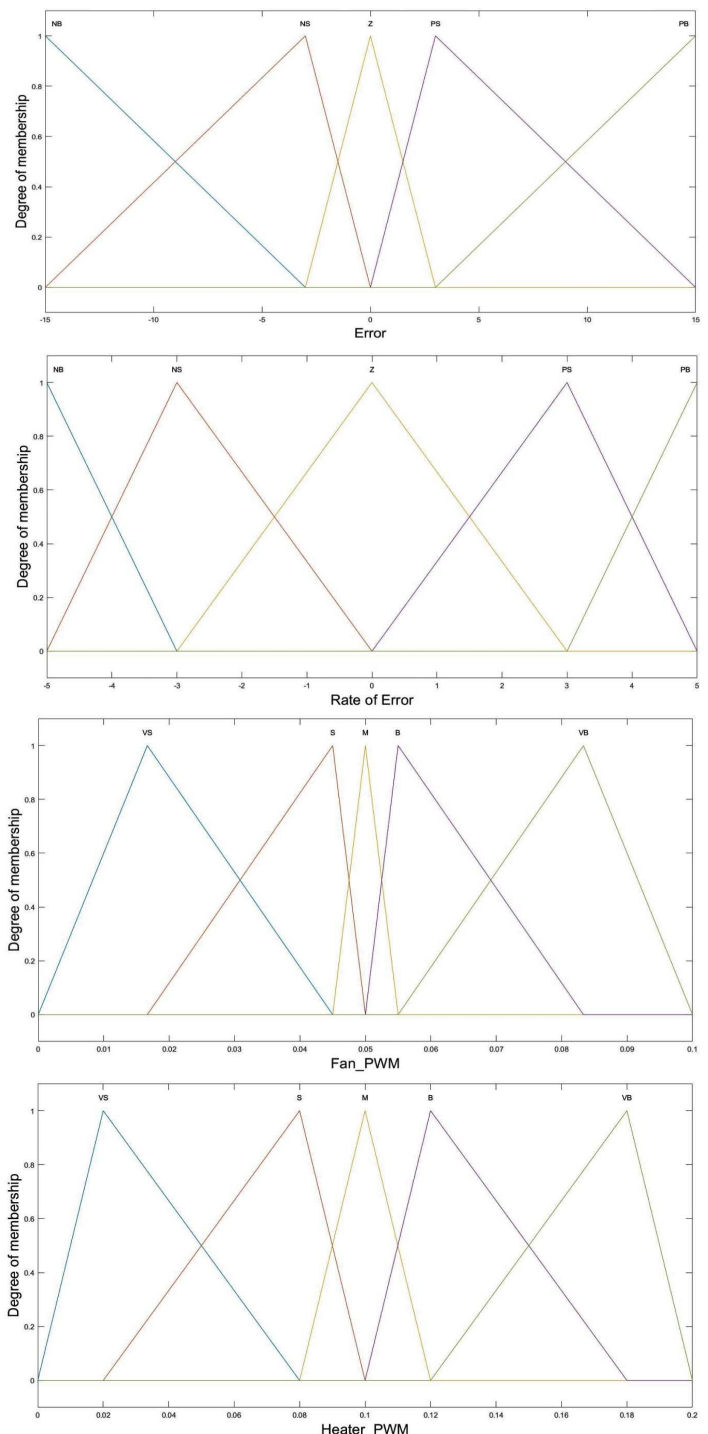


Fig. 6. Input and output membership function temperature controllers

TABLE V. SOIL MOISTURE'S VALUE WHEN PUMP IS ON

$\begin{matrix} \text{Time that pump is on} \\ \text{Time} \end{matrix}$	3s	5s	10s
0s	21.72	21.69	21.47
30s	23.94	36.53	32.51
60s	28.62	38.41	44.23
90s	35.73	41.14	53.46
120s	35.91	42.83	59.3
150s	37.87	43.58	62.40
180s	38.03	43.72	63.67

of the fuzzy controller respectively which error difference between the greenhouse humidity and the optimal humidity. Piezoelectric is the output of the fuzzy controller. Fig. 9, Fig. 10 represents membership functions of humidity inputs, and the membership functions of output shown in Fig. 11. The fuzzy rules comprehensive in Table VI. In Table VII change of humidity's value when the piezoelectric is on and also, the proposed tables contain the time that piezoelectric is on for tuning the humidity.

D. Fire alarm system

In this section, in order to protect the greenhouse from fire and dangerous gases, we use a fire alarm system. The manufactured fire alarm system is shown in Fig. 12. This system measures the greenhouse gases and if these gases are more than allowed, the system will start beeping and turn red lights. This system is also possible for users can monitor the value of gases and be notified of the occurrence of fire. Main components of the proposed system the Esp8266 Wi-Fi module, gas sensor, buzzer, and red light.

E. Greenhouse User Application

In this section is possibility provide to monitor and change the desire value of greenhouse parameters. The web user page represents in Fig. 13.

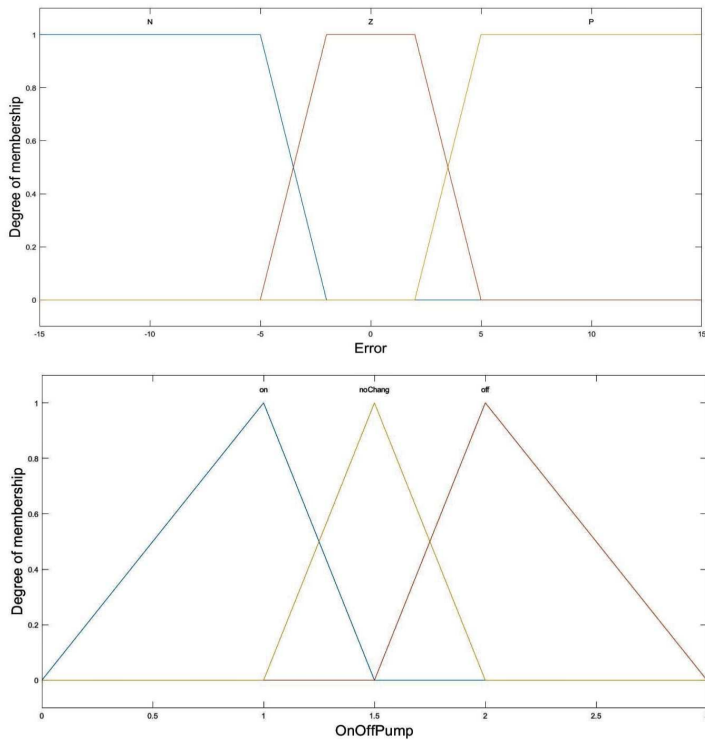


Fig. 7. The membership functions of the soil moisture fuzzy system



Fig. 8. Humidity control system

IV. PRACTICAL RESULTS

In this part, control efficiency of greenhouse parameters is investigated. In the temperature control system desired temperature for growing the greenhouse product is 25°C . In this system possibility is provided that user can monitor and

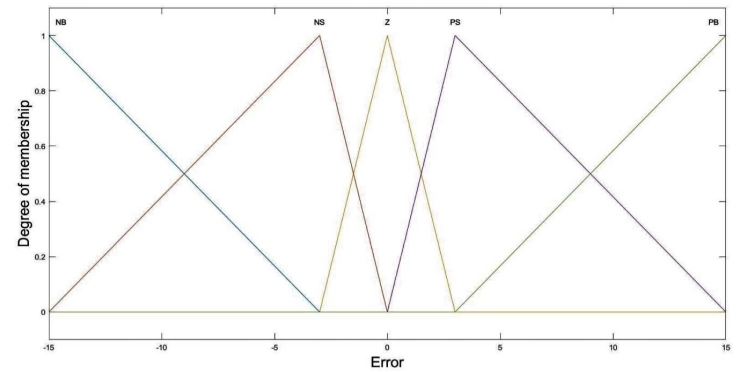


Fig. 9. First input membership functions of the humidity

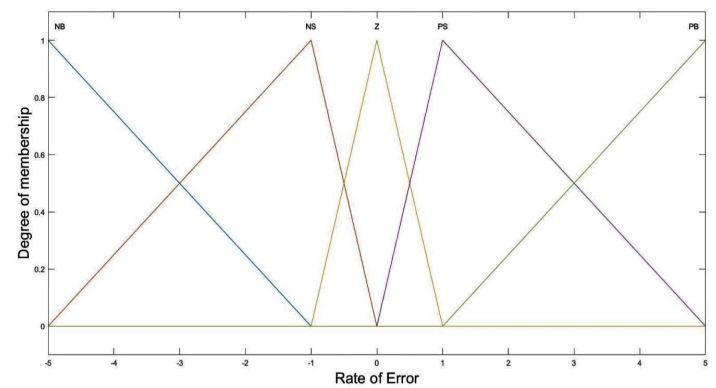


Fig. 10. Second input membership functions of the humidity

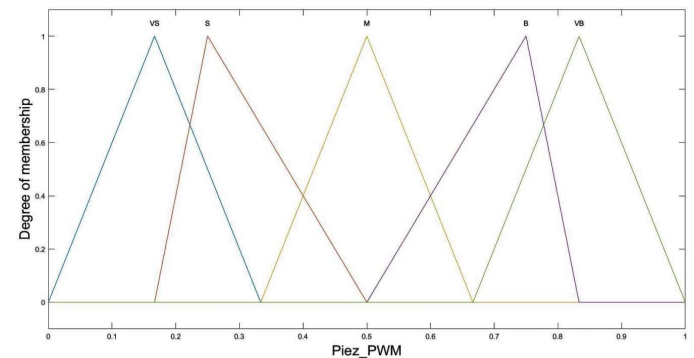


Fig. 11. Output membership functions of the humidity



Fig. 12. Fire alarm system

adjust the temperature. Entered disturbance is controlled and prevent damage to the plant. In Fig. 14 temperature control of greenhouse for 4 hours is shown. The supervisor system can capable to prevent damage to the greenhouse product.

In the control system of soil moisture desired value for the growth of the plant is 50%. In this system possibility is provided that user can monitor and adjust the soil moisture and controller is prevent damage to the plant. In Fig. 15. soil moisture control of greenhouse for 2 hours is shown.

In the control system of humidity desired value for growth of the plant is 60%. In this system possibility is provided that user can monitor and adjust the humidity and controller is prevent damage to the plant. In Fig.16 humidity control of greenhouse for 4 hours is shown.

V. CONCLUSION

This paper presented the design and implementation smart greenhouse system. The proposed procedure benefits from the advantages of the so called disruptive IoT technology. An optimization technique has also been used to come up with an optimized solution. In order to be more practical, as it is the case in the real physical system, an alarming system has also been introduced and checked on the system. The achieved results show that the proposed system is successful to be used in a practical application.

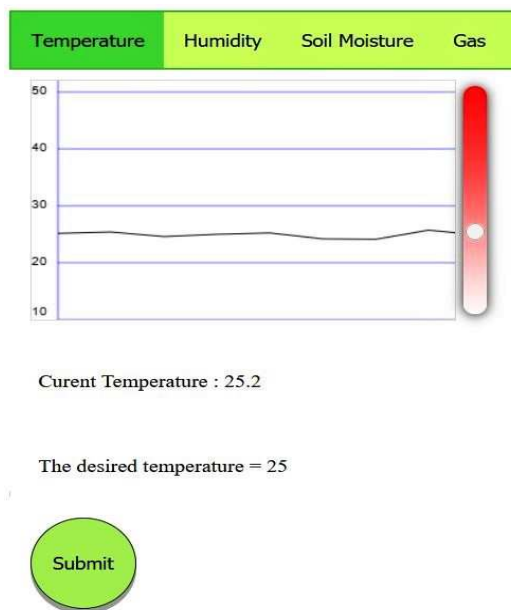


Fig. 13. Smart greenhouse user page

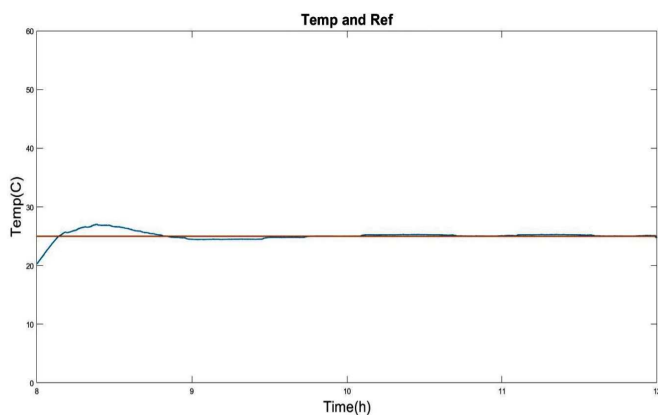


Fig. 14. Temperature of smart greenhouse

TABLE VI. CONTROL LAWS OF HUMIDITY FUZZY SYSTEM

$e \backslash \dot{e}$	NB	NS	Z	PS	PB
NB	M	B	VB	B	M
NS	S	M	B	M	S
Z	VS	S	M	S	VS
PS	S	M	B	M	S
PB	M	B	VB	B	M

TABLE VII. ENVIRONMENT HUMIDITY

Time \ pwm	10%	25%	45%	65%	80%
0s	40%	41%	40%	40%	41%
30s	46%	51%	55%	58%	60%
60s	53%	60%	66%	69%	71%
90s	55%	66%	71%	74%	79%
120s	59%	70%	78%	79%	84%
150s	62%	73%	83%	85%	90%
180s	65%	75%	86%	90%	92%

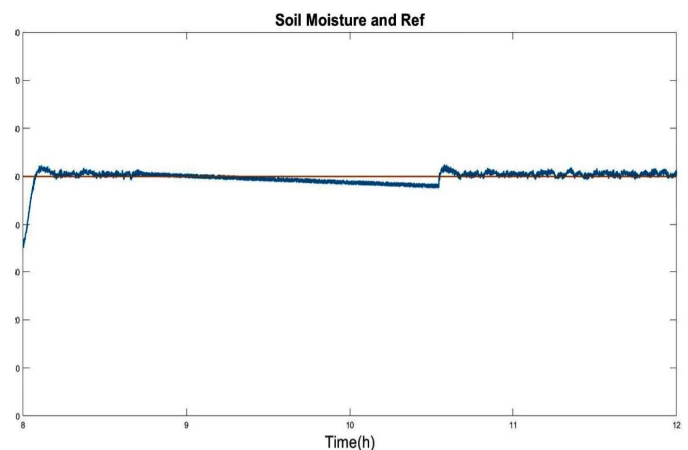


Fig. 15. Soil moisture of smart greenhouse

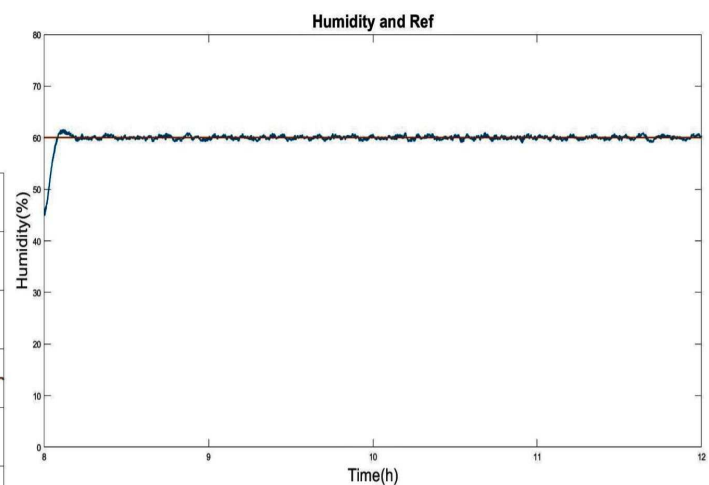


Fig. 16. Humidity of smart greenhouse

REFERENCES

- [1] Hoque, Mohammad Asadul, and Chad Davidson. "Design and implementation of an IoT-based smart home security system." *International Journal of Networked and Distributed Computing* 7, no. 2: 85-92, (2019).
- [2] Khan, Fazeel Ahmed, Adamu Abubakar Ibrahim, and Akram M. Zeki. "Environmental monitoring and disease detection of plants in smart greenhouse using internet of things." *Journal of Physics Communications* 4, no. 5: 055008, (2020).
- [3] Ullah, Mohammad Woli, Mohammad Golam Mortuza, Md Humayun Kabir, Zia Uddin Ahmed, Sovan Kumar Dey Supta, Partho Das, and Syed Mohammad Didar Hossain. "Internet of Things Based Smart Greenhouse: Remote Monitoring and Automatic Control." *DEStech Transactions on Environment, Energy and Earth Sciences iceee* , (2018).
- [4] Abioye, Emmanuel Abiodun, Mohamad Shukri Zainal Abidin, Muhammad Naveed Aman, Mohd Saiful Azimi Mahmud, and Salinda Buyamin. "A model predictive controller for precision irrigation using discrete lagurre networks." *Computers and Electronics in Agriculture* 181: 105953, (2021).
- [5] Chaudhary, Gopal, Surinder Kaur, Bhawna Mehta, and Rachna Tewani. "Observer based fuzzy and PID controlled smart greenhouse." *Journal of Statistics and Management Systems* 22, no. 2: 393-401,(2019).
- [6] Van Beveren, P. J. M., J. Bontsema, G. Van Straten, and E. J. Van Henten. "Optimal control of greenhouse climate using minimal energy and grower defined bounds." *Applied Energy* 159: 509-519, (2015).
- [7] Zhang, Lingxian, Zanyu Xu, Dan Xu, Juncheng Ma, Yingyi Chen, and Zetian Fu. "Growth monitoring of greenhouse lettuce based on a convolutional neural network." *Horticulture Research* 7, no. 1: 1-12 , (2020).
- [8] Ayu Widiana, Siska, Suryono Suryono, and Budi Warsito. "Plant Seeds Growth Prediction on Greenhouse Using Adaptive Neuro Fuzzy Inference System (ANFIS) Method." In *E3S Web of Conferences*, vol. 202, p. 14008, (2020).
- [9] Shenan, Zaidon Faisal, Ali Fadhil Marhoon, and Abbas A. Jasim. "IoT based intelligent greenhouse monitoring and control system." *Basrah Journal for Engineering Sciences* 1, no. 17 (2017): 61-69.
- [10] Y. Alaviyan., M. H. Aghaseyedabdollah, M. H. Sadafi, and A. Yazdizade. "Design and Manufacture of a Smart Greenhouse with Supervisory Control of Environmental Parameters Using Fuzzy Inference Controller." In *2020 6th Iranian Conference on Signal Processing and Intelligent Systems (ICSPIS)*, pp. 1-6. IEEE, (2020).
- [11] Keswani, Bright, Ambarish G. Mohapatra, Amarjeet Mohanty, Ashish Khanna, Joel JPC Rodrigues, Deepak Gupta, and Victor Hugo C. De Albuquerque. "Adapting weather conditions based IoT enabled smart irrigation technique in precision agriculture mechanisms." *Neural Computing and Applications* 31, no. 1: 277-292, (2019).
- [12] Nawandar, Neha K., and Vishal R. Satpute. "IoT based low cost and intelligent module for smart irrigation system." *Computers and electronics in agriculture* 162: 979-990, (2019).
- [13] Alaviyan, Y., and Ali A. Afzalian. "Decentralized Fuzzy Sliding Mode Control Using Jaya Algorithm." In *2020 6th Iranian Conference on Signal Processing and Intelligent Systems (ICSPIS)*, pp. 1-5. IEEE, (2020).
- [14] R. Venkata Rao, "Jaya: A simple and new optimization algorithm for solving constrained and unconstrained optimization problems," *International Journal of Industrial Engineering Computations*, vol. 7, pp. 19–34, 01, (2016).